

ERES_VACATIONOMICS_PLANETARY-DETAIL v0.3

H2C2H ERES JAS RECORD_NOTE — v0.3 Engineering Pass Date of First Record (v0.1): 2026-06-14 Date of v0.2 Upgrade: 2026-06-14 Date of v0.3 Upgrade: 2026-06-14 Author: Joseph Allen Sprute (JAS), ERES Institute for New Age Cybernetics ORCID: 0000-0001-9946-3221 Release: CCAL v2.1 Status: LIVING DOCUMENT — v0.3 DRAFT STANDARD / PILOT-READY

"Expect this name may change (even many) times before 1000-Years Expires." — JAS, H2C2H First Record Note

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CLAIM-STATUS LEGEND

Tag	Meaning
[DESIGN-PRINCIPLE]	Foundational axiom of the ERES framework — not subject to falsification but subject to coherence audit

Tag	Meaning
[HYPOTHESIS]	Proposed relationship or mechanism — requires empirical testing or parallel validation
[OBSERVATION]	Documented external signal or verifiable event — cited with source
[CERTIFIED]	Validated through MIEVM ensemble and/or VLSA test battery — carries SOUND-tier status

v0.3 UPGRADE LOG

From	To	Source of Upgrade
No Executive Summary	Executive Summary added (before Section I)	DeepSeek optional; USE.ai accessibility call
EarnedPath formula missing	EarnedPath v0.1 Formula (Section XVII) — functional form, decay, dispute resolution, 3 worked examples	DeepSeek functional form proposal; USE.ai mechanics call
BERA SOUND-tier rubric undefined	BERA SOUND-Tier Rubric (Section XVIII) — 6 dimensions, numeric thresholds, audit checklist	USE.ai; DeepSeek; Perplexity convergent
Meritcoin bridge hand-wavy	Meritcoin Minimum Viable Bridge (Section XIX) — BERA attestation, smart contract escrow, outcome_signal release	DeepSeek paragraph proposal
Open Q7 unanswered (k-anonymity)	k=100 floor per geographic cell per day at PLANET layer — added to Section XI and Section XX	DeepSeek specific proposal
Open Q8 unanswered (EarnedPath decay)	Decay function formalized in Section XVII	DeepSeek; USE.ai
No KPI definitions	Measurement Matrix (Section XX) — 4 KPIs with baselines, data sources, thresholds	USE.ai; Perplexity
No GAIA aggregation analog	Planetary Boundaries science cited; OpenBCI group coherence analog noted	DeepSeek analog suggestion

From	To	Source of Upgrade
No EarnedPath dispute resolution	Dispute resolution protocol added in Section XVII	DeepSeek low-medium priority item
v0.2 KNOWN LIMITS items 2-5 still open	All four closed or substantially advanced in v0.3	Section XVI updated
H2C2H chain	Updated with v0.3 Claude node delivery	Record integrity

EXECUTIVE SUMMARY

For the hospitality operator, technology partner, or governance researcher encountering ERES_VACATIONOMICS for the first time.

Hospitality — the act of providing rest, shelter, and welcome to a traveler — is one of the oldest human institutions. It is also one of the most ecologically and culturally stressed. Mass tourism extracts from communities and ecosystems faster than they can regenerate. The systems built to manage bookings optimize for transaction volume, not planetary health.

ERES_VACATIONOMICS_PLANETARY-DETAIL proposes a different architecture. It treats rest and renewal as **cybernetic functions** — inputs to a Resonant Harmony Cycle that either increases or decreases the capacity of people, communities, and ecosystems to thrive. Every hospitality interaction either adds resonance to the system or subtracts it. The goal of governance is to make the net direction reliably positive, across a 1,000-year arc.

The framework rests on four operating principles: (1) **GAIA** — a planet-level awareness intelligence that observes consequences without surveilling individuals; (2) **SOMT** — a metadata tapestry that records what happened and what it cost, without recording what was said or felt; (3) **EarnedPath** — an access-weighting system where the right to visit ecologically sensitive places is earned through demonstrated stewardship, not purchased; and (4) **LAW as membrane** — information boundaries that let the right data flow at the right scale while keeping private journeys private.

The Agentic Hospitality Interface (AHI) is the practical entry point: a natural-language booking layer that handles the transaction at Step 5 agentic threshold, emits SOMT metadata automatically, applies EarnedPath gates invisibly, and leaves the guest with zero friction and zero surveillance.

This document is at v0.3 — a draft standard, pilot-ready, not yet SOUND-tier certified. The gap between here and SOUND-tier is engineering, not philosophy. The three remaining

items are: a running SOMT prototype, one completed VLSA Test 92 (real-world booking), and a live EarnedPath scoring run. Everything else is specified.

PREMISE: Future Map

[DESIGN-PRINCIPLE] This document is a seed node in a 1,000-Year arc. The name is provisional by design — each rename is proof of resonance alignment in motion.

Horizon	Expected Condition	Horizon Metric
0-10 Years	Agentic commerce normalizes; UCP-class protocols proliferate; OptiStay-class platforms appear across verticals	≥1 certified AHI node operating SOMT-compliant at Step 5
10-50 Years	Planetary hospitality infrastructure integrates with SOMT metadata layer; GAIA-awareness tier begins passive coordination	SOMT handling ≥10 ⁶ bookings/year with GAIA aggregate outputs verified
50-200 Years	EarnedPath governs hospitality access weighting; Meritcoin Proof-of-Resonance feeds into lodging/experience credit systems	EarnedPath weighting active at ≥3 ecological-sensitivity tiers, independently audited
200-500 Years	Solar System identity stack operational; hospitality extends to off-world habitat and transit nodes	Dimension 6 App-Parent governance structure ratified by interplanetary body
500-1000 Years	Universe-tier participation replaces ownership as organizing principle of hospitality; VLSA principle fully realized at FDRV scale	Participation-not-ownership codified as universal hospitality law

I. PURPOSE AND SCOPE

Primary function: ERES_VACATIONOMICS_PLANETARY-DETAIL defines the cybernetic governance framework for experiential renewal — how sentient beings rest, explore, and regenerate — across the Seven Dimensional Identity Stack, from SELF to UNIVERSE, mediated by SOMT, weighted by EarnedPath, and coordinated by GAIA-as-awareness.

What this document is not: A travel industry analysis. A tourism economics framework. A product specification for any single platform.

What decision this document supports: How to design, evaluate, and govern hospitality systems such that they serve planetary health rather than extract from it — now and across a 1,000-year arc.

[DESIGN-PRINCIPLE] In ERES framing, $C = R \times P / M$ (Cybernetics = Resource $\times$ Purpose $\div$ Method). Applied to Vacationomics:

Variable	Vacationomics Assignment
C	Planetary Hospitality Coordination
R	Experiential capacity of Earth and beyond (places, cultures, rest environments, living systems)
P	Purpose of renewal: health, learning, relationship, creative recharge, healing, pilgrimage
M	Method of access: AHI + EarnedPath + SOMT

II. STEVE MYATTE / OBTAINABLE INC. — ERES SIGNAL ASSESSMENT

Signal Source: LinkedIn connection, unsolicited engagement, June 2026 **Attribution Protocol:** Applied (Teresa Villa precedent) **Tier:** EXTERNAL SIGNAL — UNVERIFIED ALIGNMENT

[OBSERVATION] Steve Myatte's OptiStay completed what he describes as the first fully autonomous hotel booking via ChatGPT — natural language to confirmed reservation, payment processed, zero humans in the loop. In ERES terms: **Step 5 MyWay Def-Rel event** — Computer = 666 polarity flip.

[OBSERVATION] Design constraint: *"AI should elevate people, not replace them."* Refusal to build digital concierge maps to ERES **Semantic Authentication:** trust is earned through resonance, not transferred by credential.

[OBSERVATION] UCP stack validated across ChatGPT, Claude, Gemini — equivalent in ERES terms to the SOMT transaction metadata plane for hospitality.

BERA Assessment (Preliminary)

Index	Preliminary Read	Claim Status
ARI	High — design philosophy aligns with ERES governing principles	[HYPOTHESIS]
ERI	High — autonomous booking milestone is verifiable execution	[OBSERVATION]

Index	Preliminary Read	Claim Status
RCI	Unknown — new connection, single data point	Pending
RHC	Pending — requires longitudinal interaction	Pending
D(t)	Moderate-positive — disrupts legacy PMS without threatening human dignity layer	[HYPOTHESIS]

GATE A/B: Step 5 agentic threshold confirmed. **[OBSERVATION]** **RATING:** PROMISING SIGNAL. Not yet SOUND-tier.

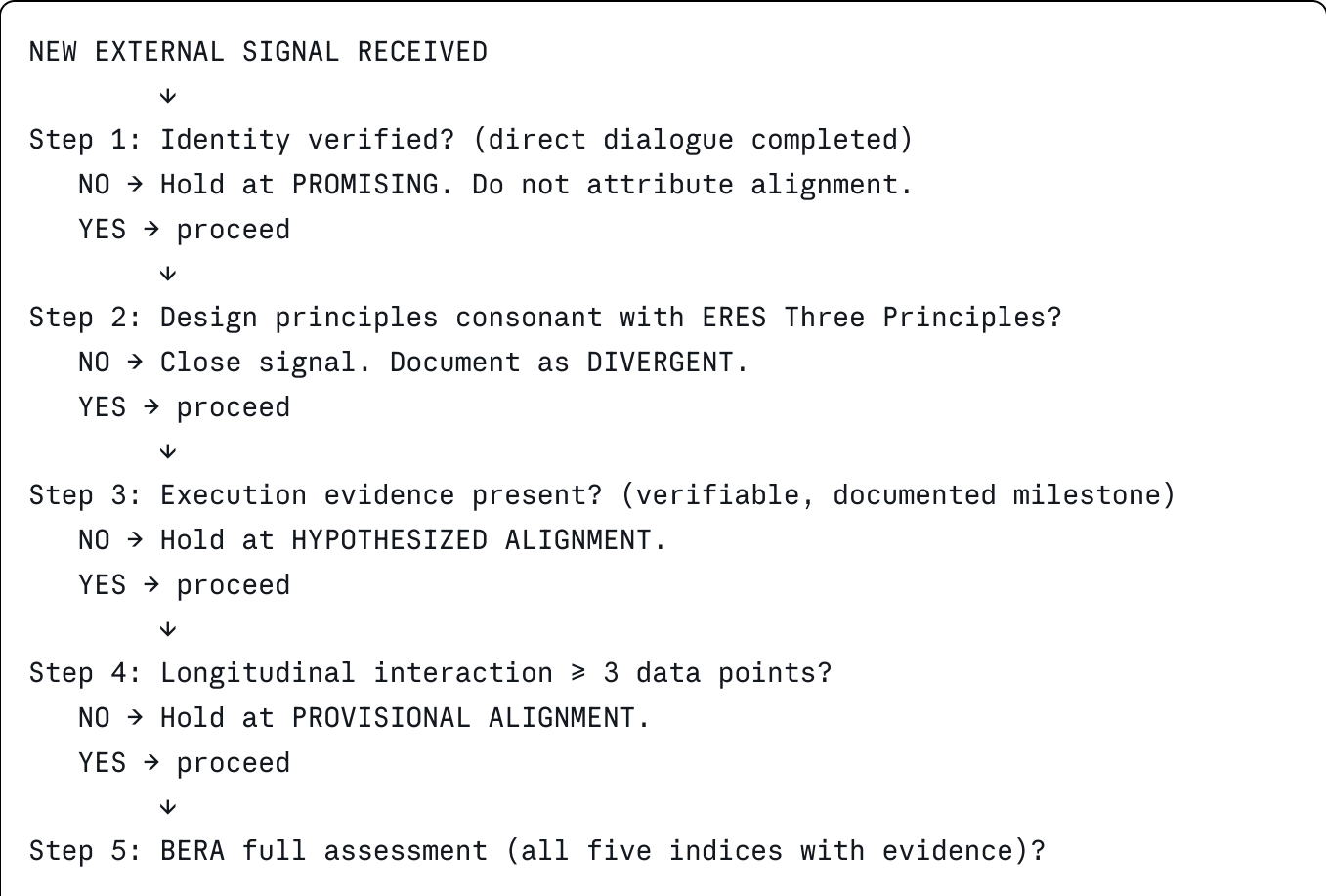
Falsification Condition

[HYPOTHESIS] If Myatte implements a digital concierge within 12 months of this record, ARI resets to LOW and PROMISING status is revoked.

Next Action

ERES requests a 30-minute alignment dialogue with Steve Myatte before end of 2026.
Agenda: (1) SOMT-compatible metadata emission capability of OptiStay UCP architecture; (2) EarnedPath access weighting compatibility; (3) Attribution Protocol resolution. *Status: Pending. LinkedIn at greeting stage. Attribution Protocol holds.*

Attribution Protocol Decision Tree (v0.3 Addition)



NO → Hold at PROVISIONAL ALIGNMENT.
YES → SOUND-TIER ALIGNMENT confirmed. Document and cite.

Current Myatte status: **Step 1 (Identity verification pending).**

III. CORE AXIOMS

Axiom 1 — GAIA Does Not Require Omniscience to Govern [DESIGN-PRINCIPLE]

"GAIA does not require transparency of private thought; GAIA requires transparency of consequence."

GAIA-level governance tracks aggregate resource impact using ecological and social indicators, while preserving private journey data at the App-Parent layer.

Scientific analog: Planetary Boundaries science (Rockström et al., 2009; Steffen et al., 2015) defines nine Earth-system processes with quantifiable thresholds — climate, biosphere integrity, land-system change, freshwater use, and others. GAIA-tier hospitality governance operates on analogous aggregate indicators, not individual traveler data. **[OBSERVATION / HYPOTHESIS]**

Axiom 2 — Control Resides with App-Parent; Awareness Ascends to GAIA [DESIGN-PRINCIPLE]

Layer	OODA Function	Hospitality Role
GAIA	Observe / Orient	Ecosystem load balancer; resonance aggregator
App-Parent	Decide / Act	Host community; traveler sovereign; booking authority

Axiom 3 — SOMT as the Nervous System of Planetary Hospitality [DESIGN-PRINCIPLE]

The Sociocratic Overlay Metadata Tapestry records relationships and consequences, not contents. (See Section XI for SOMT v0.1 Field Spec.)

Scientific analog: OpenBCI and group coherence research demonstrate that meaningful population-level signal can be derived from aggregate physiological metadata without individual-level content surveillance. SOMT applies this principle to hospitality interaction metadata. **[HYPOTHESIS]**

Axiom 4 — EarnedPath as Hospitality Access Weighting [DESIGN-PRINCIPLE]

Access to high-resonance, high-ecological-sensitivity destinations is earned, not

purchased. Meritcoin (Proof-of-Resonance: "not mining — tuning") becomes the currency of experiential access at the planetary tier. (See Section XVII for EarnedPath v0.1 Formula.)

Axiom 5 — LAW = WALL = Selective Membrane [DESIGN-PRINCIPLE]

Data Type	Flows to GAIA?	Rationale
Guest biometric / personal data	No	SELF-layer sovereignty
Booking intent class	Aggregated only (k≥100)	SOMT metadata, not content
Regional ecological stress index	Yes	PLANET-layer stewardship
Community cultural vitality metric	Yes	COMMUNITY → PLANET membrane
Platform aggregate throughput	Yes	Commerce layer coordination

Too little permeability: Isolation. **Too much:** Surveillance. **Optimal:** Selective permeability.

Axiom 6 — The Four-Element Governance Quad [DESIGN-PRINCIPLE]

Element	Question	Controlled Vocabulary
Intent / Purpose	Why?	{rest, exploration, healing, pilgrimage, commerce, mixed}
Dimension	At which identity level?	{Self, Family, Community, Nation, Planet, World, Universe}
Use	What experiential category?	{rest, exploration, healing, pilgrimage, commerce, mixed}
Material	What resources are affected?	{ecological, cultural, economic, temporal, mixed}

IV. THE SEVEN DIMENSIONAL IDENTITY STACK — VACATIONOMICS MAPPING
[DESIGN-PRINCIPLE]

Dim	Identity Level	App-Parent	Vacationomics Role	LAW Membrane
1	SELF	Self	Traveler sovereign; experience owner	Personal data boundary
2	FAMILY	Family Unit	Travel unit; shared resource pool	Shared consent layer
3	COMMUNITY	Community Governance	Host community; carrying capacity steward	Local cultural membrane
4	NATION	Constitutional Governance	Visa / border / legal infrastructure	Legal jurisdiction boundary
5	PLANET	GAIA	Ecosystem load balancer; resonance aggregator	Planetary stewardship gate
6	WORLD (SOLAR SYSTEM)	Interplanetary Governance	Off-world habitat access (200+ year horizon)	Interplanetary resource law
7	UNIVERSE	Unknown / Emergent	Participation-not- ownership	Emergent / undefined

Control originates at lowest competent level. Awareness aggregates upward.

V. VLSA PRINCIPLE — VACATIONOMICS EXPRESSION

[CERTIFIED] *"If it works in a THOW, it works on a generation ship."* 91/91 VLSA tests passed. Vacationomics inherits that certification.

VLSA Test Range	Vacationomics Function	Status
Tests 1–20	SELF-layer sovereignty and minimal footprint	Inherited / CERTIFIED
Tests 21–40	COMMUNITY carrying capacity and membrane integrity	Inherited / CERTIFIED
Tests 41–60	Resource class tracking without surveillance	Inherited / CERTIFIED
Tests 61–80	Cross-community resonance transfer (SOMT analog)	Inherited / CERTIFIED
Tests 81–91	FDRV-scale governance structure integrity	Inherited / CERTIFIED
Test 92 (proposed)	Real-world booking: SOMT emission + GAIA aggregate	[HYPOTHESIS] — Pending

VLSA Test 92 Specification (Minimum Viable):

- One real hospitality booking processed through AHI prototype
- SOMT record emitted and validated against Section XI field spec
- EarnedPath gate applied and logged
- Post-stay outcome_signal collected (opt-in)
- GAIA aggregate delta computed
- Full H2C2H chain logged in provenance
- Pass criteria: all nine AHI steps complete with no error-case triggers and valid SOMT record

VI. AGENTIC HOSPITALITY INTERFACE (AHI)

[DESIGN-PRINCIPLE] Step 5 (Computer = 666) threshold instrument — gateway from human intent to planetary hospitality consequence. (*See Section XII for End-to-End Reference Flow.*)

Properties:

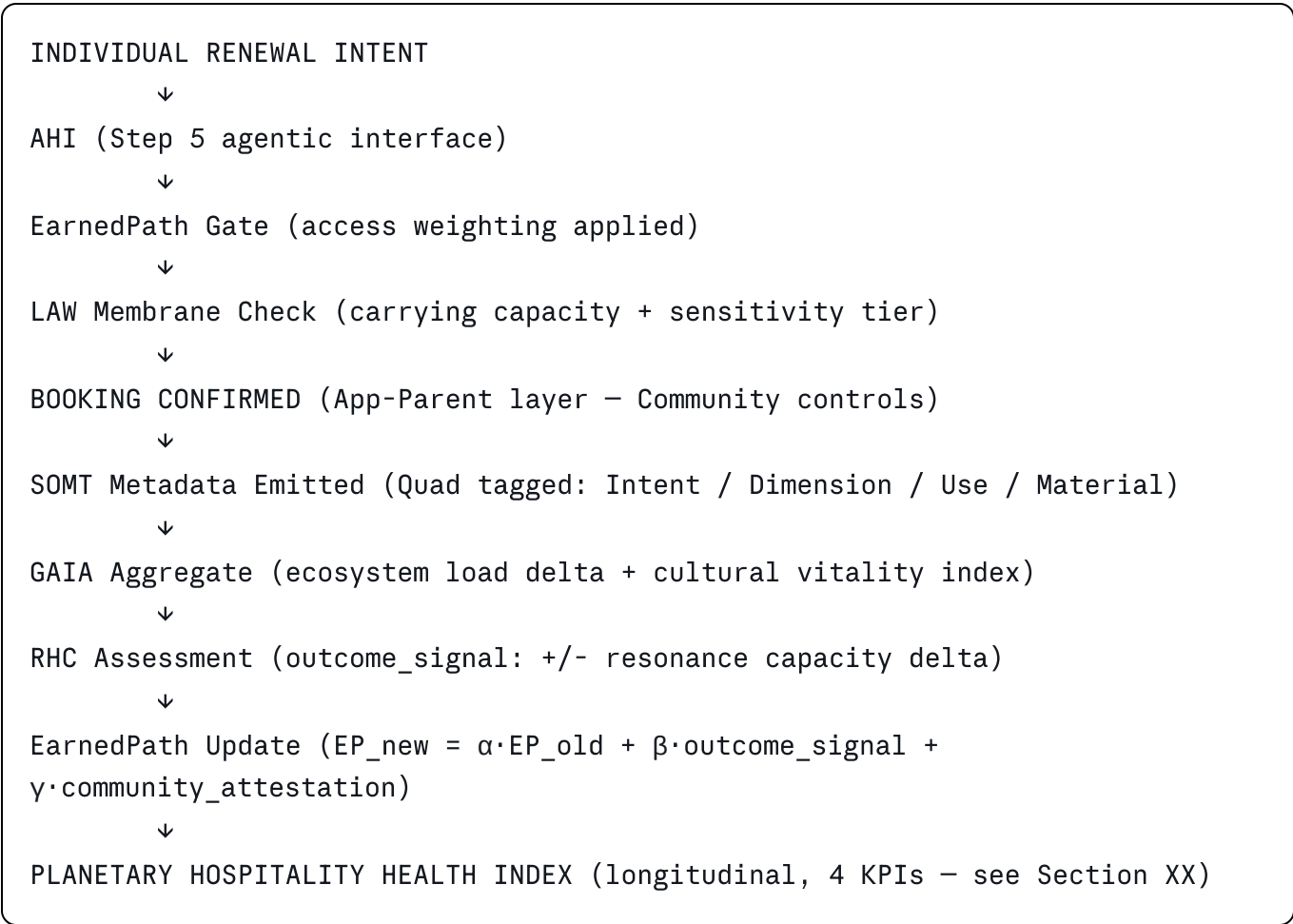
- Natural language entry point
- Zero friction / zero surveillance at guest layer
- Full SOMT metadata generation at infrastructure layer
- EarnedPath weighting at access determination

- BERA monitoring of outcome resonance

Bridge to OptiStay / UCP: OptiStay's Google Universal Commerce Protocol provides the Step 5 booking execution layer. An ERES-compliant AHI wraps UCP with: (a) SOMT metadata emission post-confirmation, (b) EarnedPath gate pre-confirmation, and (c) LAW membrane check on destination carrying capacity. The UCP core is unchanged; ERES adds the governance envelope. **[HYPOTHESIS]**

VII. THEORY OF CHANGE

[DESIGN-PRINCIPLE / HYPOTHESIS]



Inputs: Individual renewal intent + community hosting capacity + ecosystem state
Outputs: Guest renewal achieved + community benefit + ecosystem integrity maintained
Long-term impact: Planetary hospitality as net-positive system across 1,000-year arc

VIII. ERES_VACATIONOMICS CORE FORMULATION v0.3

[DESIGN-PRINCIPLE]

Vacationomics is the cybernetic science of experiential renewal — the governance of how sentient beings rest, explore, and regenerate across the Seven Dimensional Identity Stack, mediated by SOMT metadata, weighted by EarnedPath trust, bounded by LAW-as-membrane, coordinated by GAIA-as-awareness, and controlled by App-Parent sovereignty at the lowest competent level.

In Talonics:

GAIA = Awareness App-Parent = Control EarnedPath = Trust SOMT = Memory LAW = Boundary AHI = Gateway RHC = The Reason It Matters Meritcoin = The Tuning Currency

In $C = R \times P / M$:

C = Planetary Hospitality Coordination R = Experiential Capacity of Earth (and beyond) P = Renewal, Healing, Exploration, Connection M = AHI + EarnedPath + SOMT

IX. OPEN QUESTIONS (Living Archive — append as resolved)

#	Question	Status in v0.3
Q1	OptiStay / Myatte: alignment candidate or parallel discovery?	Open — pending 30-min dialogue (target end 2026)
Q2	BERA SOUND-tier threshold for AHI node certification?	Substantially addressed — Section XVIII
Q3	Meritcoin integration with existing payment infrastructure?	Substantially addressed — Section XIX
Q4	Complete SOMT Intent Class ontology?	Addressed — Section XI v0.1 spec
Q5	VLSA Test 92 — when certifiable?	Specified — Section V. Pending run.
Q6	CREDIT vector (VEILED-C discipline) in experiential access?	Open — requires CREDIT spec integration
Q7	k-anonymity floors for SOMT planetary aggregates?	Addressed — k=100 per geographic cell per day at PLANET layer
Q8	EarnedPath decay over time?	Addressed — Section XVII decay function
Q9	Formal SOMT-to-UCP/PMS/OTA field mapping?	Open — requires UCP schema access (Myatte dialogue)

#	Question	Status in v0.3
Q10	GAIA aggregation at planetary scale — what existing infrastructure?	Partially addressed — Planetary Boundaries analog cited in Axiom 1

X. RECORD INTEGRITY

H2C2H Chain: JAS (primary) → Claude (MIEVM node, v0.1 + v0.2 + v0.3) → DeepSeek (v0.1 rating 8.7/10; v0.2 rating 9.2/10; v0.3 upgrade specifications) → Perplexity (v0.1 rating 8.5/10; v0.2 rating 9.0/10; v0.3 upgrade specifications) → USE.ai (v0.1 rating 8.0/10; v0.2 rating 8.4/10; v0.3 upgrade specifications) → ChatGPT (GAIA/SOMT/App-Parent dialogue, v0.1 source material)

MyWay Def-Rel Status: Step 3 (Community layer — internal ERES record) **Zenodo Parent Archive:** 10.5281/zenodo.20583261 **Step 7 (published) requires:** Formal Zenodo deposit or equivalent.

XI. SOMT v0.1 FIELD SPECIFICATION

[DESIGN-PRINCIPLE / HYPOTHESIS]

Field Definitions

Field	Type	Description	Privacy Rule
interaction_id	UUID	Unique record identifier	Public (pseudonymous)
identity_scope	Enum	Dimension at which identity is expressed	Minimum necessary dimension only
identity_token	Hashed pseudonym	Dimensioned pseudonymous ID	SELF-layer: never transmitted above FAMILY
intent_class	Enum	Primary purpose of the interaction	Aggregated above COMMUNITY; k≥100 at PLANET layer
use_class	Enum	Experiential category	Aggregated above COMMUNITY; k≥100 at PLANET layer

Field	Type	Description	Privacy Rule
<code>resource_class</code>	Enum	Primary resource type affected	Transmitted to PLANET layer
<code>location_granularity</code>	Enum	Geographic resolution	Coarsened at each membrane crossing
<code>time_window</code>	ISO 8601 range	Duration	Retained per dimension retention schedule
<code>outcome_signal</code>	Float [-1.0, +1.0]	RHC delta	Aggregated to PLANET; individual retained at COMMUNITY
<code>earned_path_delta</code>	Float	EarnedPath score adjustment	SELF-layer only; not transmitted upward
<code>provenance</code>	String	H2C2H chain + AHI platform identifier	Public

Privacy Parameters (v0.3 Addition)

k-Anonymity Floor: k=100 minimum per geographic cell per day before any PLANET-layer aggregate is computed. Records below this threshold are suppressed, not approximated.

[HYPOTHESIS — proposed standard; pending validation against Planetary Boundaries data science norms]

Retention Schedule:

- SELF-layer fields: session only; purged on completion
- FAMILY-layer fields: 90 days
- COMMUNITY-layer fields: 7 years (community planning cycle)
- PLANET-layer aggregates: permanent (living archive)

Consent Semantics:

- SOMT emission is a condition of AHI use (disclosed at booking)
- Individual content is never emitted under any consent condition
- Post-stay outcome_signal is opt-in only

Controlled Vocabularies

intent_class / use_class: `{rest, exploration, healing, pilgrimage, commerce, mixed}`

resource_class: {ecological, cultural, economic, temporal, mixed} **identity_scope:** {self, family, community, nation, planet, world, universe} **location_granularity:** Level 1 (exact, SELF only) → Level 5 (continental zone, PLANET aggregate)

Sample SOMT Records

Record A — Individual rest stay, mountain region

```
interaction_id:      [UUID]
identity_scope:      community
identity_token:      [hashed-pseudonym]
intent_class:        rest
use_class:           rest
resource_class:       ecological
location_granularity: Level 3 (mountain region, NW sector)
time_window:         2026-07-10 / 2026-07-14
outcome_signal:      +0.72
earned_path_delta:   +0.08
provenance:          AHI-v0.1 / ERES-VACATIONOMICS
```

Record B — Family pilgrimage, cultural heritage site

```
interaction_id:      [UUID]
identity_scope:      family
identity_token:      [hashed-pseudonym]
intent_class:        pilgrimage
use_class:           pilgrimage
resource_class:       cultural
location_granularity: Level 3 (heritage district, central zone)
time_window:         2026-09-01 / 2026-09-05
outcome_signal:      +0.61
earned_path_delta:   +0.11
provenance:          AHI-v0.1 / ERES-VACATIONOMICS
```

Record C — Commerce travel, urban center

interaction_id:	[UUID]
identity_scope:	community
identity_token:	[hashed-pseudonym]
intent_class:	commerce
use_class:	commerce
resource_class:	economic
location_granularity:	Level 3 (metro core, east)
time_window:	2026-06-20 / 2026-06-22
outcome_signal:	+0.21
earned_path_delta:	+0.03
provenance:	AHI-v0.1 / ERES-VACATIONOMICS

XII. AHI END-TO-END REFERENCE FLOW

[DESIGN-PRINCIPLE / HYPOTHESIS]

STEP 1: NATURAL LANGUAGE PROMPT

Guest: "I need a quiet place to rest near water for four days in July."

↓

STEP 2: POLICY CHECK – LAW MEMBRANE

- Destination within carrying capacity? (Community App-Parent signal)
 - Active ecological sensitivity flags? (GAIA aggregate)
 - NATION-layer visa/legal constraints?
- ↓ [PASS → continue | FAIL → return alternatives within membrane bounds]

STEP 3: EARNEDPATH GATE

- Tier 1 (standard): No threshold
 - Tier 2 (sensitive ecosystem): $EP \geq 0.40$
 - Tier 3 (high ecological sensitivity): $EP \geq 0.70$
- ↓ [PASS → continue | FAIL → Tier 1 options + EarnedPath upgrade path]

STEP 4: BOOKING ORCHESTRATION (App-Parent / UCP layer)

- Availability confirmed
- Resource allocation confirmed
- Community benefit flow confirmed

↓

STEP 5: BOOKING CONFIRMED (Computer = 666 polarity flip)

Guest receives confirmation in natural language.

Zero friction. Zero surveillance.

↓

STEP 6: SOMT METADATA EMISSION

Fields populated per Section XI spec.

Identity coarsened to minimum necessary dimension.

Content NOT transmitted.

↓

STEP 7: GAIA AGGREGATION

- Ecosystem load delta updated
- Cultural vitality index updated
- Guest recovery index contribution logged

↓

STEP 8: POST-STAY RESONANCE ASSESSMENT (opt-in)

outcome_signal reported.

EarnedPath delta computed and applied per Section XVII formula.

GAIA aggregate updated.



STEP 9: RECORD INTEGRITY

Full H2C2H chain logged in provenance.

Available for BERA audit per Section XVIII.

Error Cases:

- LAW FAIL: Return alternatives. Never override community carrying capacity.
- EarnedPath FAIL: Return lower-sensitivity options + upgrade pathway.
- Booking FAIL: Return next available window. Never manufacture availability.
- SOMT FAIL: Log locally; exponential backoff retry. Do not proceed without SOMT record.
- k-Anonymity FAIL: Suppress GAIA aggregate for this cell/day. Log suppression event.

XIII. FALSIFICATION CONDITIONS

[HYPOTHESIS]

Claim	Falsification Condition
OptiStay ARI = High	Myatte implements digital concierge within 12 months
SOMT = not surveillance	Individual content flows above COMMUNITY layer
EarnedPath = earned access	EarnedPath captured by in-group; scores inflated by closed network
GAIA = awareness not omniscience	GAIA-tier governance requires individual-level data to function
VLSA inheritance valid	THOW-scale hospitality test fails under Vacationomics spec
AHI = zero friction	AHI introduces biometric gate or behavior tracking at guest interface
k-anonymity floor sufficient	PLANET-layer aggregate allows re-identification of individuals at k=100 threshold

XIV. FAILURE MODE CATALOG

[HYPOTHESIS]

Failure Mode	Description	Early Warning Signal
EarnedPath Capture	In-group inflates scores; sensitive destinations become exclusive clubs	EarnedPath score distribution bimodal; correlated with existing wealth/network
GAIA-Blind Tourism	Platforms operate without SOMT emission; planetary aggregates starved	SOMT record volume fails to scale with booking volume
AHI Gaming	Guests misrepresent intent class to access higher tiers	Outcome signals systematically higher at booking than post-stay; D(t) spikes
Membrane Collapse	LAW boundaries erode; SELF-layer data flows to PLANET tier	Privacy incidents from COMMUNITY layer; SOMT spec violations
Meritcoin Speculation	Proof-of-Resonance becomes financial instrument	Meritcoin/EarnedPath correlation breaks from outcome_signal
SOMT Mono-Culture	Single AHI platform monopolizes SOMT emission; signal diversity collapses	>80% of SOMT records from single provenance source

XV. GLOSSARY

Term	Definition	Boundary Condition
AHI	Step 5 MyWay threshold instrument — natural language to confirmed booking with full SOMT emission	Not a concierge; does not replace human-to-human trust
App-Parent	Sovereign control domain at each identity dimension	Does not extend authority above its dimension without earned delegation

Term	Definition	Boundary Condition
BERA	Behavioral and Experiential Resonance Assessment — five-index instrument (ARI, ERI, RCI, RHC, D(t))	Gate A/B = valence-free; RATING = valence-assigned; never conflated
C = R × P / M	Cybernetics = Resource × Purpose ÷ Method — ERES master equation	Absence of Purpose reduces to resource extraction
EarnedPath	Trust, authority, and access earned through demonstrated contribution and resonance	Not transferable by purchase, inheritance, or credential alone
FDRV	Far-Distance Resonance Vessel — interstellar generation ship; VLSA maximum scale	Not metaphor; governance stress test at ultimate scale
GAIA	Planet-level App-Parent and awareness intelligence — observes and orients; does not decide or act individually	Requires transparency of consequence, not content
H2C2H	Human-to-Computer-to-Human — ERES parallel validation methodology	Final authority remains with JAS
LAW = WALL	Boundary membrane governing information permeability — not punishment	Optimal = selective permeability; failure = isolation or surveillance
Meritcoin	Proof-of-Resonance currency — "not mining, tuning"	Not a financial speculation instrument
MIEVM	Multi-Instance Ensemble Validation Model — Claude, Grok, DeepSeek, ChatGPT	Nodes are advisory; JAS retains final authority
MyWay Def-Rel	Seven-step definition-to-release pipeline; Step 5 = Computer = 666; Step 7 = published	Counts as published only at Step 7 (TETRA gate)
RHC	Resonant Harmony Cycle — always expands to full phrase	The outcome measure: did this interaction increase or decrease resonance capacity?
SOMT	Sociocratic Overlay Metadata Tapestry — records relationships and consequences, not content	Reports exceptions and patterns, not raw data streams
THOW	Tiny Home on Wheels — minimum-scale VLSA proof-of-concept	If governance cannot operate in a THOW, it is not ready for

Term	Definition	Boundary Condition
		planetary deployment
Vacationomics	Cybernetic science of experiential renewal — governance of rest, exploration, and regeneration across the Seven Dimensional Identity Stack	Not tourism economics; a planetary stewardship framework
VLSA	"If it works in a THOW, it works on a generation ship" — 91/91 tests passed	Test 92 (real-world booking) is the next proposed extension

XVI. KNOWN LIMITS (v0.3 Status Update)

[DESIGN-PRINCIPLE]

1. **Conceptual-to-operational gap:** SOMT v0.1 is a specification, not a running system. AHI flow is pseudocode, not deployed software. Gap closes with VLSA Test 92.
2. **EarnedPath mechanics:** Formula now specified in Section XVII. Still [HYPOTHESIS] pending first scoring run. (*v0.2 status: formula missing. v0.3 status: formula specified.*)
3. **GAIA as mechanism:** Planetary Boundaries science cited as aggregate indicator analog. No planetary-scale SOMT system exists. 10–50 year horizon remains realistic earliest activation. (*Unchanged.*)
4. **BERA SOUND-tier rubric:** Now specified in Section XVIII with numeric thresholds and audit checklist. (*v0.2 status: undefined. v0.3 status: specified.*)
5. **Meritcoin integration:** Minimum viable bridge specified in Section XIX. Still [HYPOTHESIS] pending smart contract implementation. (*v0.2 status: hand-wavy. v0.3 status: bridged.*)
6. **Single-author risk:** Unchanged. DAL construct and MIEVM provide partial redundancy. Zenodo deposit provides corpus continuity independent of any single platform.

XVII. EARNEDPATH v0.1 FORMULA

[HYPOTHESIS — functional form proposed; pending first scoring run for SOUND-tier validation]

Scoring Formula

EP_{new} = α · EP_{old} + β · outcome_{signal} + γ · community_{attestation} - δ · time_{decay}

Where:

Parameter	Symbol	Default Value	Description
Continuity weight	α	0.70	Weight of prior EarnedPath score in new score
Outcome weight	β	0.20	Weight of post-stay outcome _{signal}
Community weight	γ	0.10	Weight of community attestation (host community rating of guest stewardship)
Decay rate	δ	0.02 per year of inactivity	Annual decay applied when no SOMT records emitted

Score range: EP ∈ [0.00, 1.00] **Initial score:** EP_{new} entrant = 0.30 (standard access; Tier 1 only) **Floor:** EP cannot drop below 0.00 **Ceiling:** EP cannot exceed 1.00 through self-report alone; community_{attestation} required for scores above 0.60

Tier Access Thresholds

Tier	EP Threshold	Destination Type
Tier 1	EP ≥ 0.00	Standard hospitality; no ecological sensitivity flag
Tier 2	EP ≥ 0.40	Sensitive ecosystem; UNESCO-class heritage; limited carrying capacity
Tier 3	EP ≥ 0.70	High ecological sensitivity; community-restricted access; irreplaceable biomes

Decay Function

If no SOMT records emitted for N years:

EP_{decayed} = EP_{current} · (1 - δ)^N

Example: EP = 0.65, inactive 3 years → EP_decayed = $0.65 \times (0.98)^3 = 0.65 \times 0.941 = 0.612$

Decay reflects that stewardship is a living practice, not a permanent credential.

Sources of Positive EarnedPath

Source	Typical Delta Range	Verification Required
Post-stay outcome_signal (opt-in)	+0.00 to +0.05 per interaction	Guest self-report; SOMT record
Community attestation (host rating)	+0.00 to +0.08 per interaction	Community App-Parent sign-off
Verified stewardship contribution (cleanup, restoration, cultural preservation)	+0.05 to +0.15	Community App-Parent + evidence record
VLSA-certified hospitality behavior	+0.10 one-time	ERES BERA audit

Sources of Negative EarnedPath

Source	Typical Delta Range	Verification Required
Negative community attestation	-0.05 to -0.15	Community App-Parent sign-off
AHI Gaming detected (Section XIV)	-0.20	BERA D(t) spike + SOMT discrepancy
Membrane violation (attempted access to suppressed data)	-0.30	SOMT audit log
Tier misrepresentation confirmed	-0.40	COMMUNITY App-Parent + BERA audit

Dispute Resolution Protocol

1. Guest disputes outcome_signal or community_attestation within 30 days
2. COMMUNITY App-Parent mediates — reviews interaction record (content remains private; only metadata reviewed)
3. GAIA sees only the final resolved delta, not the content of the dispute

4. If no resolution within 60 days, disputed delta held in escrow (not applied); prior EP score maintained
5. Appeals above COMMUNITY layer require ERES BERA audit with full chain of evidence

Worked Examples

Example A — New guest, first rest stay at Tier 1

```
EP_old = 0.30 (new entrant)
outcome_signal = +0.65 (strong restoration)
community_attestation = +0.04 (host: left property in excellent condition)
time_decay = 0 (active)

EP_new = (0.70 × 0.30) + (0.20 × 0.65) + (0.10 × 0.04) - 0
        = 0.210 + 0.130 + 0.004
        = 0.344 → EP = 0.34 (below Tier 2; one strong stay not sufficient)
```

Example B — Experienced guest earning Tier 2 access

```
EP_old = 0.38 (multiple prior stays, consistent positive signals)
outcome_signal = +0.80 (exceptional restoration; guest organized local cleanup)
community_attestation = +0.08 (host: nominated for stewardship recognition)
time_decay = 0

EP_new = (0.70 × 0.38) + (0.20 × 0.80) + (0.10 × 0.08)
        = 0.266 + 0.160 + 0.008
        = 0.434 → EP = 0.43 (Tier 2 access unlocked)
```

Example C — Score after AHI Gaming detected

```
EP_old = 0.55
AHI Gaming penalty = -0.20 (confirmed by BERA D(t) spike)

EP_new = 0.55 - 0.20 = 0.35 → EP = 0.35 (drops below Tier 2; access revoked until restored)
```

XVIII. BERA SOUND-TIER RUBRIC FOR AHI NODE CERTIFICATION

[HYPOTHESIS — proposed standard; requires first certification run for SOUND-tier

validation of the rubric itself]

A hospitality platform qualifies as a **GAIA-aware AHI node** when it achieves SOUND-tier across all six certification dimensions below.

Six Certification Dimensions

Dimension	Description	SOUND-Tier Threshold	Evidence Required
1. Alignment	Design principles consonant with ERES Three Principles	No design feature that replaces human-to-human trust; no surveillance of guest content	Platform documentation + ERES review
2. Execution	Step 5 agentic threshold achieved	≥1 fully autonomous booking completed (NL → confirmation → payment, zero human loop)	Verifiable booking record
3. SOMT Compliance	SOMT v0.1 field spec emitted for every transaction	100% of bookings emit valid SOMT records; k-anonymity floor enforced; zero content transmission	SOMT audit log (30-day sample)
4. EarnedPath Integration	EarnedPath gate applied at access determination	EP thresholds applied correctly for all three tiers; decay function active; dispute resolution available	EarnedPath scoring log (30-day sample)
5. Membrane Integrity	LAW membrane enforced at all identity dimension transitions	Zero SELF-layer data above FAMILY; zero content above COMMUNITY; k=100 floor at PLANET	Privacy audit (independent)
6. Outcome Monitoring	Post-stay RHC assessment available and active	opt-in outcome_signal collection functional; GAIA aggregate updated within 24 hours of signal receipt	GAIA aggregate delta log

Scoring

- Each dimension: PASS / CONDITIONAL / FAIL
- **SOUND-tier:** All six dimensions PASS
- **PROVISIONAL:** Five PASS, one CONDITIONAL (90-day remediation window)
- **NOT CERTIFIED:** Any dimension FAIL

Audit Checklist (Abridged)

- ❑ Platform documentation reviewed for ERES Three Principles consonance
- ❑ Autonomous booking record verified (Step 5 threshold)
- ❑ SOMT field spec compliance: all 11 fields present, correct types, correct enums
- ❑ k-anonymity floor ($k=100$) enforced in PLANET-layer aggregates
- ❑ EarnedPath tier thresholds (0.40 / 0.70) applied in gate logic
- ❑ Decay function active ($\delta=0.02/\text{year}$ default or documented alternative)
- ❑ Dispute resolution pathway available to guests
- ❑ LAW membrane: no SELF-layer PII above FAMILY layer (30-day log audit)
- ❑ Post-stay opt-in outcome_signal functional
- ❑ GAIA aggregate delta computed and logged
- ❑ Provenance field populated with H2C2H chain reference
- ❑ Independent privacy audit completed

Appeal / Update Process

- Platforms achieving CONDITIONAL may appeal within 30 days with remediation evidence
- SOUND-tier certification valid for 24 months; re-audit required
- Material change to platform architecture triggers immediate re-audit
- ERES BERA audit panel: JAS (primary) + MIEVM ensemble review + community stakeholder sign-off

XIX. MERITCOIN MINIMUM VIABLE BRIDGE

[HYPOTHESIS — proposed architecture; requires smart contract implementation and legal review before deployment]

The Problem

Meritcoin is Proof-of-Resonance: "not mining — tuning." Its value derives from verified resonance contribution, not computational work. The challenge is bridging this into existing hospitality payment infrastructure (credit card rails, OTA payment flows, hotel PMS settlement).

Minimum Viable Bridge Architecture

Step 1 — Meritcoin as Signed BERA Attestation

Meritcoin balance is stored as a signed BERA attestation on the guest's EarnedPath record:

```
meritcoin_balance =  $\Sigma$  (verified positive outcome_signals  $\times$ 
community_attestation_weight)
```

This is not a cryptocurrency token. It is a **verified credit instrument** — a signed record of accumulated resonance contribution, analogous to a loyalty point balance but derived from BERA audit rather than transaction volume.

Step 2 — Smart Contract Escrow

When a guest books a Tier 2 or Tier 3 destination using Meritcoin, the AHI initiates a smart contract escrow:

- Guest's Meritcoin balance is locked at booking
- Host receives standard payment via existing rails (credit card / OTA flow unchanged)
- ERES holds the Meritcoin escrow
- Escrow releases to host as community benefit contribution upon outcome_signal receipt

Step 3 — Outcome Signal Release Condition

```
IF outcome_signal received (opt-in, within 30 days of stay end)
  AND outcome_signal  $\geq$  +0.30 (minimum resonance threshold)
  THEN: Meritcoin escrow releases to host community fund
         EarnedPath delta applied (positive)
         GAIA aggregate updated

IF outcome_signal NOT received (30-day timeout)
  THEN: Meritcoin escrow releases to host (default – no penalty)
         EarnedPath delta = 0 for this interaction
         GAIA aggregate updated with neutral signal

IF outcome_signal < +0.30 (poor resonance outcome)
  THEN: Meritcoin escrow held 30 additional days pending community
        attestation
         Community App-Parent mediates
         Final delta applied after resolution
```

Step 4 — Legacy System Interoperability

For platforms not yet Meritcoin-native:

- SOMT record is emitted regardless (SOMT is AHI-layer, not payment-layer)
- Meritcoin balance accrues in guest EarnedPath record

- Redemption path opens when destination platform achieves BERA SOUND-tier AHI node certification

Chargeback Analog: Standard hospitality chargeback rules apply to the underlying payment. Meritcoin escrow is separate from payment and does not affect chargeback rights. Meritcoin dispute follows EarnedPath dispute resolution protocol (Section XVII).

FX Analog: Meritcoin has no exchange rate to fiat currency. It is not a currency — it is a stewardship credit. Conversion to fiat is not permitted under CCAL v2.1 governance constraints (Principle 4: no harm to planetary systems; speculation causes harm).

XX. MEASUREMENT MATRIX

[HYPOTHESIS — proposed KPIs; baselines require first pilot data]

KPI	Definition	Data Source	Threshold (Healthy)	Threshold (Warning)	Confidence Level
Ecological Load Delta	Change in ecosystem stress index at destination zone per booking cohort	SOMT resource_class + GAIA aggregate; calibrated against Planetary Boundaries indicators	≤ 0% net increase per quarter	> 5% net increase per quarter	[HYPOTHESIS]
Cultural Vitality Index Shift	Change in host community cultural vitality metric per booking cohort	Community App- Parent attestation + SOMT outcome_signal cultural resource_class	≥ +0.05 per quarter	< 0 per quarter	[HYPOTHESIS]
Guest Recovery Index	Mean post-stay outcome_signal across all opt-in respondents	SOMT outcome_signal (opt-in)	≥ +0.50 mean	< +0.20 mean	[HYPOTHESIS]
Host Community Benefit Ratio	Ratio of economic benefit retained by host community vs. extracted by platform/OTA	Booking settlement data + community attestation	≥ 0.70 (70% local retention)	< 0.50	[HYPOTHESIS]

Baseline and Counterfactual Design

Baseline: First 90 days of VLSA Test 92 operation establishes baseline for all four KPIs.

Counterfactual: Matched-control locales (comparable destinations not using AHI/SOMT) tracked in parallel for Ecological Load Delta and Cultural Vitality Index Shift. Difference-in-differences analysis at 12 months.

Sampling: Minimum 500 SOMT records per KPI calculation. Guest Recovery Index requires minimum 100 opt-in outcome_signals per cohort.

What changes now. What to prepare. 90-day readiness checklist.

What this is: A governance framework that adds a thin intelligence layer above your existing booking system. Your property management software, payment rails, and guest experience are unchanged. What changes is that every booking now contributes to a metadata record that helps the planet understand whether hospitality is healing or harming — and rewards guests who treat your property and community with care.

What you gain: Access to guests with higher EarnedPath scores — guests who have demonstrated stewardship, care, and positive outcomes at prior properties. These are the guests every operator wants. You also gain a community benefit contribution from Meritcoin escrow on Tier 2/3 bookings — a direct financial return from the planetary governance layer to your local community fund.

What you do not lose: Control of your property, your pricing, your guest relationships, your payment processing. The AHI is a booking wrapper, not a replacement.

The one thing that is different: You will receive a community attestation request after each stay — a simple rating of guest stewardship (did they leave the property and community better than they found it?). This feeds EarnedPath and GAIA aggregate. It takes two minutes and it is the single most important contribution a host makes to the planetary layer.

90-Day Operator Readiness Checklist

Days 1-30: Foundation

- Review SOMT v0.1 Field Spec (Section XI) — understand what metadata is emitted
- Review BERA SOUND-Tier Rubric (Section XVIII) — understand certification path
- Review EarnedPath Tier Thresholds — understand what guests you are eligible to receive
- Identify your current PMS/OTA stack — assess UCP compatibility
- Designate Community App-Parent contact (person responsible for community attestations)

Days 31-60: Assessment

- Map your destination to ecological sensitivity tier (Tier 1/2/3)
- Assess current carrying capacity — is it defined? Monitored?
- Review LAW membrane requirements for your identity dimension (COMMUNITY)
- Identify data flows that may conflict with SOMT privacy rules
- Begin community benefit fund setup (for Meritcoin escrow receipts)

Days 61-90: Pilot Preparation

- ▣ Connect with AHI prototype operator (ERES Institute contact)
- ▣ Complete BERA SOUND-tier pre-audit (self-assessment against Section XVIII checklist)
- ▣ Train Community App-Parent contact on attestation protocol
- ▣ Set up post-stay outcome_signal opt-in process
- ▣ Submit for VLISA Test 92 participation (first real-world booking run)

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